SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ECA0908

DIGTAL SIGNAL PROCESSING

ERROR ANALYSIS TASK(CO3-AT3)

Question-2

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A DSP program uses floor() instead of round() during quantization. For the samples [1.78, 2.62, –1.37, –2.81], identify the incorrect outputs and debug the code.

- The floor() function always rounds a number down to the next lower integer.
- The round() function rounds to the nearest integer.

Input	floor()	round()	Status
1.78	1	2	❌ Incorrect
2.62	2	3	❌ Incorrect
−1.37	−2	−1	❌ Incorrect
−2.81	−3	−3	✅ Correct

Code using floor() function

```c
#include <stdio.h>
#include <math.h>

int main()
{
    float samples[] = {1.78, 2.62, -1.37,
-2.81};
    int n = 4;
    int i;

    printf("Input\t\tFloor()\t\tRound()\n");

printf("--------------------------------------\n
");

    for(i = 0; i < n; i++)
    {
        printf("%.2f\t\t%.0f\t\t%.0f\n",
               samples[i],
               floor(samples[i]),
               round(samples[i]));
    }

    return 0;
}
```

Debugged code(using round())

```c
#include <stdio.h>
#include <math.h>

int main()
{
    float samples[] = {1.78, 2.62, -1.37,
-2.81};
    int n = 4;
    int i;

    printf("Correct Quantized Outputs:\n");

    for(i = 0; i < n; i++)
    {
        printf("%.2f --> %.0f\n", samples[i],
round(samples[i]));
    }

    return 0;
}
```